

Roll No.....

MID SEMESTER EXAMINATION MARCH-2025

Name of the Program: B. Tech. Biotech

Semester: 2nd

Name of the Course: Advanced Organic Chemistry

Course Code: TCH-202

Time: 90 minutes

Maximum Marks: 50

Note:

(i) Answer all questions by choosing any one of the sub questions

(ii) Each question carries 10 marks

Q 1	10 marks	CO-2
(a)	Discuss the Nitration reaction of benzene with mechanism	
OR		
(b)	Elaborate the effect of substituent on further electrophilic substitution reaction in benzene	
Q 2	10 marks	CO-2
(a)	Explain the term electrophiles and carbenes with suitable example	
OR		
(b)	Evaluate the SN^2 reaction with mechanism and explain which alkyl halide give SN^2 reaction faster	
Q 3	10 marks	CO-2
(a)	Define the following terms (i) Hyperconjugation (ii) Electromeric and Mesomeric effect	
OR		
(b)	Explain the term carbanion in terms of hybridization, classification and stability	
Q 4	10 marks	CO-5
(a)	Elaborate the preparation, physical and chemical properties of glucose.	
OR		
(b)	Explain the chain lengthening process of carbohydrates	
Q 5	10 marks	CO-5
(a)	Discuss the Ruff degradation reaction.	
OR		
(b)	Provide a brief introduction to carbohydrates and classify them with suitable examples.	

